Ex no: 4	Continuous Integration/Continuous Deployment of a React Application using GitHub and Jenkins Pipeline Project
Date:	

Aim:

To automate the build, test, and deployment process of a React application using GitHub for version control and Jenkins Pipeline Project for CI/CD.

Procedure :

Step 1: Create GitHub Repository and Clone

1. Create a new repository in GitHub "**Expense Tracker"**.
2. Clone the repository into your local system inside a folder
3. Open the folder in Visual Studio Code.

Step 2: Create a New React Project

1. Open the terminal in VS Code inside the cloned repository folder.
2. Run the following command to create a React project using Vite:
   npm create vite@latest Currency-exchange
3. Choose the required framework (**React**) and variant (**JavaScript**).
4. Commit and push the project files to GitHub:
   git add .
   git commit -m "feat:Files added"
   git push origin main

Step 3: Configure Jenkins Pipeline Job

1. Open Jenkins → New Item → Select Pipeline → Name it (Trip-pipeline)
2. In Pipeline section, choose Pipeline script.
3. Paste the following pipeline script:

```
pipeline {
   agent any

   environment {
      DEPLOY_DIR =
"C:\\ReactDeploy\\Expense-
Tracker"
```

```
git branch: 'main', url:
            'https://github.com/SowndaryaRaj
            akumar/Expense-Tracker.git'
                }
            }

            stage('Install Dependencies')
            {
                steps {
                    bat 'npm install'
                }
            }

            stage('Build Project') {
                steps {
                    bat 'npm run build'
                }
            }

            stage('Deploy Project') {
                steps {
                    bat """
                    if not exist
            "%DEPLOY_DIR%" (
                        mkdir
            "%DEPLOY_DIR%"
                    )

                    xcopy /E /I /Y "dist\\*"
            "%DEPLOY_DIR%\\"
                    """
                }
            }
            stage('Start Server') {
```

```
>nul 2>nul

                start "" cmd /c "cd /d
%DEPLOY_DIR% && serve -s .
-l 8081"
                """
            }
        }
    }

    post {
        success {
            echo
'=========================
================'
            echo 'BUILD
SUCCESSFUL'
            echo 'Application URL:
http://localhost:8081'
            echo
'=========================
================'
        }

        failure {
            echo
'=========================
================'
            echo 'BUILD FAILED'
            echo
'=========================
================'
        }
    }
}
```

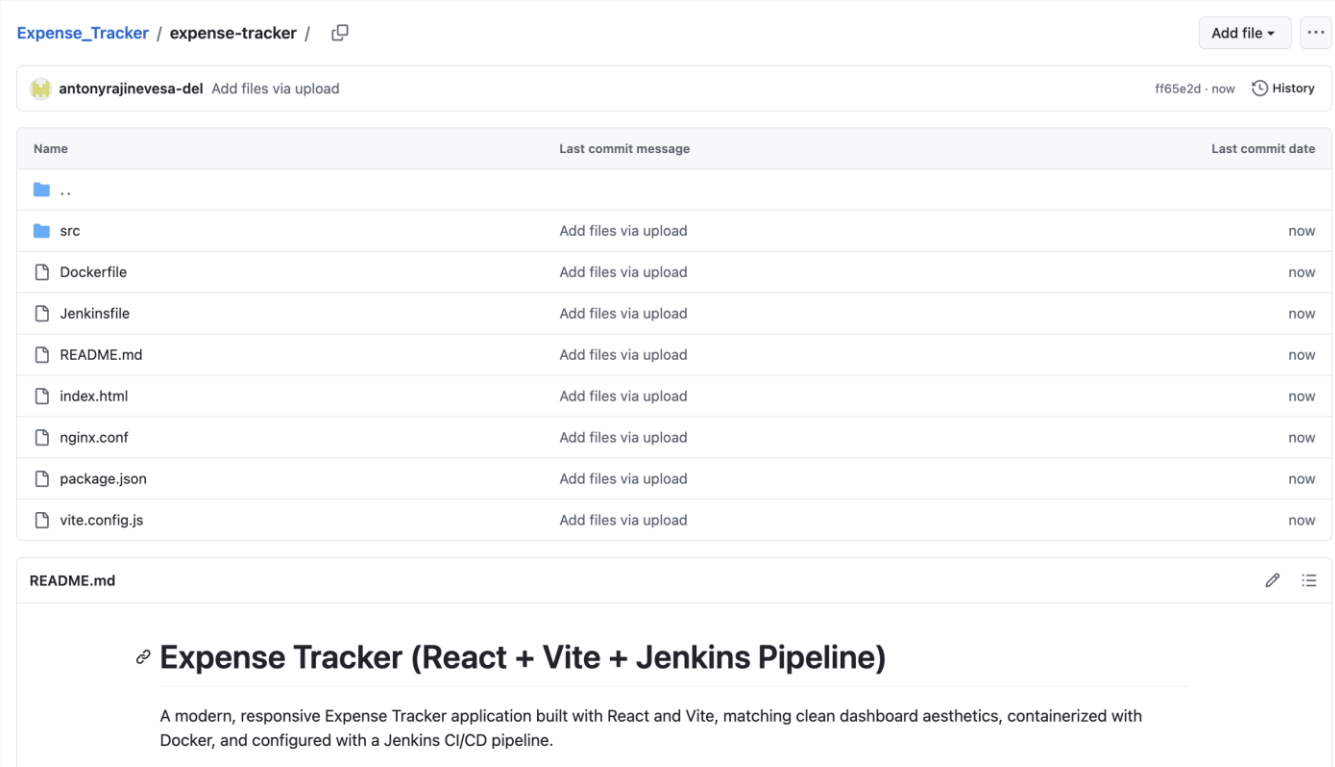
Step 4: Run the Pipeline

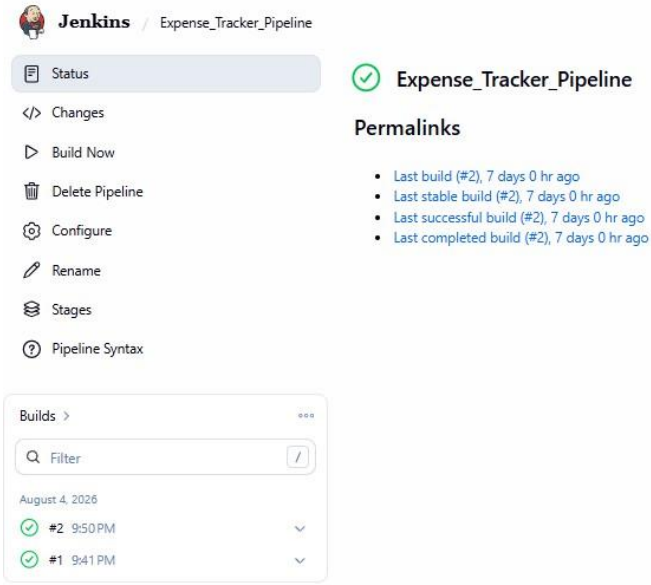
1. Save the pipeline configuration.
2. Click Build Now.
3. Jenkins will execute the stages:
    a. Checkout → pulls code from GitHub
    b. Install Dependencies → installs required node modules
    c. Build React App → builds optimized React app
    d. Deploy → copies dist folder into Jenkins userContent directory

Step 5: Run Locally with Serve

1. Open Command Prompt.
2. Navigate to the dist folder:
   cd C:\ProgramData\Jenkins.jenkins\workspace\ExpenseTracker\dist
   start serve -s . -l 8081
3. Copy the provided local link and paste it into the browser to access the app locally.

OUTPUT :


Expense_Tracker / expense-tracker /
Add file
antonyrajinevesa-del Add files via upload
ff65e2d · now
History
Name
Last commit message
Last commit date
..
src
Add files via upload
now
Dockerfile
Add files via upload
now
Jenkinsfile
Add files via upload
now
README.md
Add files via upload
now
index.html
Add files via upload
now
nginx.conf
Add files via upload
now
package.json
Add files via upload
now
vite.config.js
Add files via upload
now
README.md
Expense Tracker (React + Vite + Jenkins Pipeline)
A modern, responsive Expense Tracker application built with React and Vite, matching clean dashboard aesthetics, containerized with Docker, and configured with a Jenkins CI/CD pipeline.


Jenkins
Expense_Tracker_Pipeline
Status
Changes
Build Now
Delete Pipeline
Configure
Rename
Stages
Pipeline Syntax
Builds
Filter
August 4, 2026
#2 9:50 PM
#1 9:41 PM
Expense_Tracker_Pipeline
Permalinks
Last build (#2), 7 days 0 hr ago
Last stable build (#2), 7 days 0 hr ago
Last successful build (#2), 7 days 0 hr ago
Last completed build (#2), 7 days 0 hr ago

```
C:\expense-tracker\dist>npm install -g serve

changed 85 packages in 7s

26 packages are looking for funding
  run `npm fund` for details

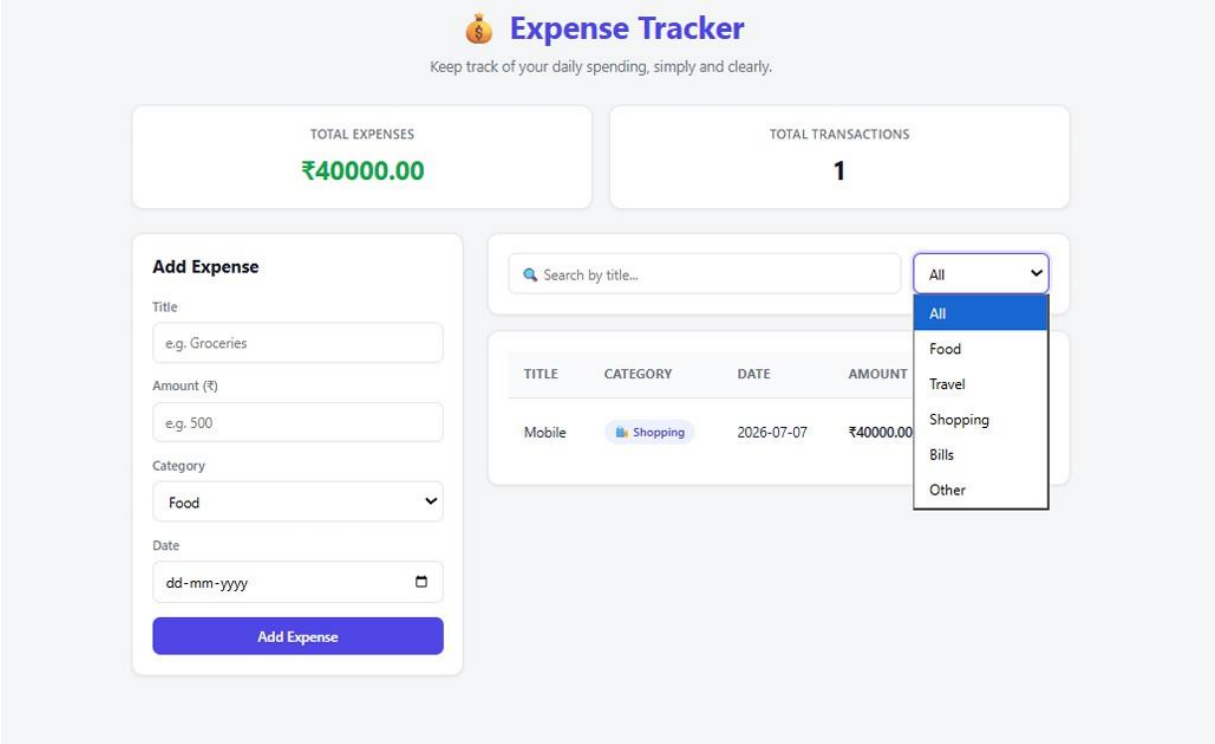
C:\expense-tracker\dist>serve -s . -l 8081

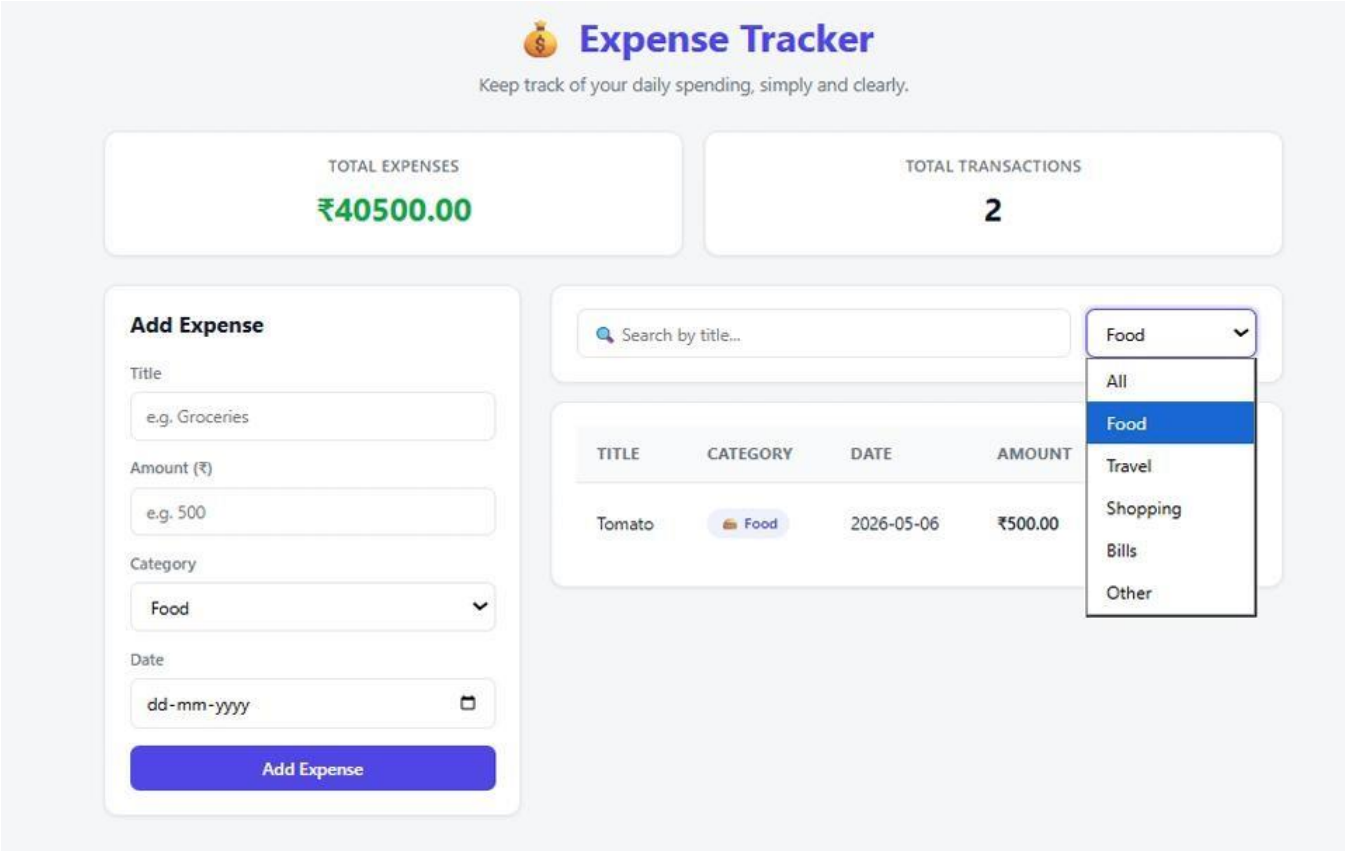
   Serving!

   - Local:    http://localhost:8081
   - Network:  http://10.11.11.195:8081

   Copied local address to clipboard!

HTTP  5/8/2026 4:11:31 pm ::1 GET /
HTTP  5/8/2026 4:11:31 pm ::1 Returned 304 in 160 ms
HTTP  5/8/2026 4:11:32 pm ::1 GET /assets/index-BlNi40-e.css
HTTP  5/8/2026 4:11:32 pm ::1 GET /assets/index-BtJMuvyd.js
HTTP  5/8/2026 4:11:32 pm ::1 Returned 304 in 11 ms
HTTP  5/8/2026 4:11:32 pm ::1 Returned 304 in 11 ms
HTTP  5/8/2026 4:11:32 pm ::1 GET /favicon.ico
HTTP  5/8/2026 4:11:32 pm ::1 Returned 304 in 4 ms
```





GitHub commands syntax & Description (10)	Implementation and Execution (15)	Tool/ Technology usage (10)	Code Quality & Documentation (5)	Time Management (5)	Viva (5)	Total (50)

Result:

The React application was successfully integrated with GitHub and Jenkins Pipeline Project. The build and deployment process was automated, and the application was successfully deployed and executed locally.